DriftDream: Training-Free Generation of Static Images with a Prescribed Illusory-Motion Field

1 Introduction

A still image can look like it is moving. Patterns in the peripheral drift family, of which Kitaoka's *Rotating Snakes* is the best known example, are printed on paper and do not change, yet most viewers see the rings turn Faubert and Herbert (1999); Kitaoka and Ashida (2003). The effect comes from repeating elements whose luminances follow an asymmetric order, together with the small involuntary eye movements that occur during fixation. Elements at different luminances are registered by the visual system at slightly different times, and a motion detector reads that timing difference as movement Conway et al. (2005); Backus and Oruç (2005). The percept is stronger in observers whose fixation is less stable, which ties it directly to eye movements rather than to the image alone Murakami et al. (2006).

Recent work has shown that a frozen text-to-image diffusion model can be made to produce illusions without any training, by splitting the denoising signal along an axis on which one canvas can satisfy two descriptions at once. Visual Anagrams uses pixel permutations, so the image shows one thing upright and another when flipped or rotated Geng et al. (2024a). Factorized Diffusion splits by spatial frequency, giving hybrid images in the sense of Oliva et al. (2006) Geng et al. (2024b). Images that Sound splits across modality, so a single canvas is both a picture and an audio spectrogram Chen et al. (2024).

Every one of these axes is instantaneous. The image is read once, and the two readings are of the same frozen array of pixels. Motion is not on the list, and the reason is not that nobody thought of it. It is that the obvious way to write the objective does not work.

Why the obvious objective is empty. Suppose we want a still image whose perceived motion matches a target field $\mathbf{F}^*$, and we have a differentiable optical-flow model M. The natural loss is to read the flow of the image and compare it to $\mathbf{F}^*$. But a still image has no flow. Feed M the same frame twice and it returns approximately zero everywhere, for every image, so the loss is constant and its gradient is zero. There is nothing to descend. The illusion we are trying to create does not live in a single frame; it appears only when the image moves slightly on the retina.

What we do instead. We put the eye movements into the objective. At each optimisation step we take the current image, apply a small set of rigid shifts that stand in for microsaccades, read the flow between consecutive shifted frames, remove the part of that flow explained by the shift itself, and average what is left over several shift directions. The average residual is the illusory component. We compare it to $\mathbf{F}^*$ with a normalised correlation and use the result as a guidance term on a frozen generator. We call this *jitter-ensemble motion guidance*. It is the one new piece of machinery in the paper, and it is what turns an empty objective into a signal that can be followed.

Two properties follow from the construction rather than from tuning. The residual is signed, so asking for clockwise and asking for anticlockwise give different images rather than the same image twice. And it is defined per pixel, so the target can be localised: rotation inside one ring, or drift confined to the shape of a letter while the surround stays still.

What we do not do. No network is trained, fine-tuned, or adapted. Every model in the pipeline is frozen and used only in the forward and backward pass. The optimised variables are the image (or the diffusion latent), nothing else.

Why the evaluation is the hard part. A model that scores an image is also a model that can be fooled. Optimising pixels against a flow network is close to the setting in which adversarial examples are found, and an image that drives M to any chosen output need not look like it moves to a person. This concern is not hypothetical for this particular percept. Rosario et al. (2026) tested a range of flow models on Rotating Snakes and found that most produce flow fields that do not match what people report; only a model built to mirror the V1–MT pathway Sun et al. (2025) showed the expected rotation, and only when the stimulus was jittered to imitate eye movements. On the predictive-coding side, Watanabe et al. (2018) reported that a PredNet trained only to predict the next frame reproduces the illusion, but Kirubeswaran and Storrs (2023) retrained the same architecture ten times and found the behaviour came and went across runs, and never appeared for greyscale versions that people still see.

We therefore separate the model that is optimised from the models that judge. Guidance uses one reader. Reporting uses readers that never appear in any gradient, plus a two-alternative forced-choice study in which people report only the direction they see. We also require a negative control: a transform that destroys the percept for human viewers must drive the measured correlation to zero. Building that control turned out to be less obvious than expected, and we describe in Section 3.7 why the first version we tried made the score go up rather than down.

Contributions.

1. A training-free method that generates a still image whose *perceived* motion follows a field the user specifies, including its direction and where in the frame it appears. Prior illusion work with diffusion priors is instantaneous; prior work on generating motion illusions Sinapayen and Watanabe (2021) controls neither content nor direction.
2. Jitter-ensemble motion guidance: an objective defined on the residual flow averaged over simulated microsaccades in several directions. We show the ensemble is required, not cosmetic — with zero jitter the measured correlation is zero by construction, and with one direction it is dominated by reader artefacts.
3. An evaluation protocol built around held-out readers, a signed control (asking for the opposite direction must reverse the measurement), and a negative control that must collapse the score. We report a case in which an intuitive control fails, and what replaces it.

2 Related Work

2.1 Illusions from frozen diffusion models

Geng et al. (2024a) generate multi-view illusions by estimating noise from several transformed views of the same noisy image and averaging the estimates before each reverse step. Their analysis shows the argument holds when the views are orthogonal transformations, which covers flips, rotations by multiples of 90°, colour inversion, and jigsaw permutations. Geng et al. (2024b) replace the transformation with a decomposition of the noise into frequency bands, recovering hybrid images Oliva et al. (2006) and related effects. Chen et al. (2024) denoise one canvas under an image model and an audio-spectrogram model at the same time, so the result can be viewed and played.

These share a structure: a fixed decomposition of the canvas, and two or more conditions imposed on the parts. Our objective is not a decomposition. There is no second prompt and no second view. There is one image, and a measurement made on it by a model that is shown the image in motion. The quantity being controlled is not what the image depicts under a transformation but what happens in an observer when the image is held still.

2.2 Guidance and search at inference time

Classifier guidance Dhariwal and Nichol (2021) and classifier-free guidance Ho and Salimans (2022) established that the sampling trajectory can be pushed toward an objective without retraining. Universal guidance Bansal et al. (2023) and self-guidance Epstein et al. (2023) extend this to arbitrary off-the-shelf networks

and to the model's own internal activations. We follow the same recipe: differentiate a loss computed on the current estimate of the clean image and add the gradient to the state.

Closest in mechanism is Motion Guidance Geng and Owens (2024), which backpropagates through a frozen flow network Teed and Deng (2020) to move content in an edit. The distinction matters and is easy to blur. Their target displacement is measured between two *different* images, a source and an output, and the content really does move. Ours is measured between an image and a shifted copy of *itself*, and nothing moves; what changes is what a motion model reports. Applying their objective to a single still frame reduces exactly to our zero-jitter ablation, which is why we report that ablation instead of implementing a separate baseline.

When gradients are unavailable or badly behaved, one can search instead. Ma et al. (2025) search over initial noises using a verifier, and Singhal et al. (2025) use interacting-particle Feynman–Kac methods to steer sampling with an arbitrary reward. Both apply to our objective without change, since it is a scalar score, and we keep them as a fallback path.

2.3 Peripheral drift and its explanations

Faubert and Herbert (1999) described the peripheral drift illusion, and Kitaoka and Ashida (2003) characterised the Rotating Snakes variant, in which motion is perceived following the repeating order black, dark, white, light. Conway et al. (2005) showed the effect survives when colour is removed, so the critical variable is the luminance relation of neighbouring elements, and recorded direction-selective cells in macaque V1 and MT that respond directionally to static element pairs. They report that high-contrast elements (black and white, relative to the mean) reach peak response 10–20 ms earlier than the intermediate greys, which supplies the timing difference a motion detector needs. Backus and Oruç (2005) give an account in terms of the time course of responses to contrast and luminance, and Fermüller et al. (2010) attribute the effect to causal temporal filtering. Murakami et al. (2006) found that observers with less stable fixation report stronger illusory motion, which is the empirical basis for building eye movements into our objective rather than treating them as noise.

2.4 Models that do or do not reproduce the illusion

Watanabe et al. (2018) showed that a PredNet Lotter et al. (2017) trained for next-frame prediction produces rotational flow on Rotating Snakes. Kirubeswaran and Storrs (2023) retrained the architecture from ten different initialisations and found the behaviour was not reliable across runs, and that no instance produced rotation for greyscale variants that people do see. Sun et al. (2025) build a model of the V1–MT pathway with a trainable motion-energy stage and a recurrent attention stage, extending an earlier version Sun et al. (2023). Rosario et al. (2026) compare several flow models on Rotating Snakes and find that most fail; the dual-channel model of Sun et al. (2025) shows the expected rotation, most clearly under simulated saccades. They also note that all models tested lack any mechanism for peripheral vision, while the human effect is stronger away from the fovea.

We take two things from this line of work and one warning. We take the simulated-saccade protocol and the normalised flow correlation as our measurement, and the target field of Rosario et al. (2026) as our default $\mathbf{F}^*$ (Section 3.2). The warning is that no single reader can carry the claim, which is why our reporting is cross-model.

2.5 Generating motion illusions

Sinapayen and Watanabe (2021) search for motion illusions with an evolutionary algorithm, scoring candidates with a PredNet, and confirm with a survey that the results work on people. That is the closest existing system, and the differences are in what can be asked for. Their search has no content term, so outputs are abstract repeating patterns rather than pictures of anything; it scores an image with a scalar, so there is no way to ask for a particular direction or to place the effect in a chosen region; and it is gradient-free evolution over a pattern-generating network rather than guidance of a generative prior. Our objective is a per-pixel signed field, which is what makes direction and layout controllable, and it attaches to a diffusion model, which is what lets the image also be a picture.

Relatedly, Gomez-Villa et al. (2019) showed that convolutional networks can be fooled by classical illusions, and Poole et al. (2023) popularised the pattern of optimising a signal through a frozen model, which we use in a different setting.

3 Method

3.1 Overview and notation

The input is a target motion field $\mathbf{F}^* : \Omega \to \mathbb{R}^2$ defined on a support mask $\Omega \subseteq \{1..H\} \times \{1..W\}$, an optional text prompt, and a frozen generator. The output is an image $\mathbf{x} \in [0,1]^{3\times H\times W}$.

Write $\mathbf{p} = (p_x, p_y)$ for a pixel with p_y increasing downwards, matching image row order. A flow field $\mathbf{f} \in \mathbb{R}^{2\times H\times W}$ has $\mathbf{f}_1$ the horizontal and $\mathbf{f}_2$ the vertical component, again with positive meaning downwards. A reader M maps a stack of frames to one flow field. We use $m(\mathbf{p}) \in [0,1]$ for the soft mask.

The pipeline has four parts, described in turn: the target field (§3.2), the simulated fixation (§3.3), the residual flow that forms the signal (§3.4), and the loss and guidance step (§3.5). Section 3.6 describes the reader we supply so that the method runs without any gated download, and Section 3.7 the controls.

3.2 Target field

For rotation we use the field of Rosario et al. (2026), who introduced it to score models against the human percept of Rotating Snakes. With centre $\mathbf{c}$, outer radius R, radial exponent γ, and $r = \|\mathbf{p} - \mathbf{c}\|$, the unit tangent and the field are

$$\mathbf{u}(\mathbf{p}) = \frac{1}{r}\begin{pmatrix} -(p_y - c_y) \\ p_x - c_x \end{pmatrix}, \qquad \tilde{\mathbf{F}}(\mathbf{p}) = \omega \left(\frac{r}{R}\right)^{\gamma} \mathbf{u}(\mathbf{p})\, m(\mathbf{p}). \tag{1}$$

The sign $\omega = \pm 1$ selects the direction and $\gamma = 1$ gives the linear radial profile of rigid rotation, avoiding a magnitude discontinuity at the outer edge. A check on the convention: at the top of the circle $p_y < c_y$ and $p_x = c_x$, so $\mathbf{u} = (1, 0)$; with p_y downwards, motion to the right at the top of a circle is clockwise, so $\omega = +1$ means clockwise on screen.

Radial expansion replaces $\mathbf{u}$ with the outward unit radial vector, and laminar drift replaces it with a constant unit vector inside a stencil, allowing targets such as "this word drifts left while the background is still".

The field is normalised over the mask,

$$\mathbf{F}^* = \tilde{\mathbf{F}} \,/\, \|\tilde{\mathbf{F}}\|_\Omega, \qquad \|\mathbf{f}\|_\Omega^2 = \sum_{\mathbf{p}} m(\mathbf{p})^2 \|\mathbf{f}(\mathbf{p})\|^2, \tag{2}$$

because the score in (6) is scale invariant: only direction and relative magnitude are being asked for, not a speed in pixels per second, which a still image does not have.

3.3 Simulated fixation

Let $\mathcal{W}_{\boldsymbol{\delta}}$ be a translation by $\boldsymbol{\delta} \in \mathbb{R}^2$ implemented by bilinear sampling,

$$(\mathcal{W}_{\boldsymbol{\delta}}\mathbf{x})(\mathbf{p}) = \mathbf{x}(\mathbf{p} - \boldsymbol{\delta}), \tag{3}$$

so image content moves by $+\boldsymbol{\delta}$. Sampling outside the frame uses reflection padding; zero padding puts a hard edge at the border and the reader reports large spurious flow there. $\mathcal{W}_{\boldsymbol{\delta}}$ is differentiable in $\mathbf{x}$, which is what lets gradients pass from the reader back to the image.

A fixation clip is a sequence of cumulative offsets $\boldsymbol{\delta}_1 = \mathbf{0}, \boldsymbol{\delta}_2, \ldots, \boldsymbol{\delta}_K$ along a straight path with step length Δ. We use J such paths with directions spread evenly over the circle, $\theta_j = 2\pi j/J$. From (3), the true flow from frame $i-1$ to frame i of path j is the constant field $\boldsymbol{\delta}_{j,i} - \boldsymbol{\delta}_{j,i-1}$.

Using several directions is not a detail. With one direction the measurement is dominated by whatever that particular shift excites in the reader: on a fixed image we measured the score swinging between $+0.13$ and -0.09 across jitter seeds. Averaging over directions removes most of that, and it also removes an easy exploit, since structure tuned to one shift direction does not survive the average.

3.4 Residual flow

The flow the reader returns is dominated by the shift we applied. The illusory part is what remains. Two ways to remove the shift are available and they do not behave the same.

The direct choice subtracts the known offset, $\mathbf{f}^{\text{res}} = M(\cdot) - (\boldsymbol{\delta}_{j,i} - \boldsymbol{\delta}_{j,i-1})$. This is only correct if the reader's gain is one. Real motion models under-report speed. On a smoothed random texture at 96×96, our built-in reader recovers a 2-pixel shift as 0.73 pixels, a gain of 0.37, so the subtraction leaves a residual proportional to the shift, pointing along the jitter, and far larger than the effect we are looking for: with $K = 4$, $\Delta = 5$ and a single direction, mean residual magnitude 4.95 pixels.

We instead remove the uniform component the reader actually reported, using the spatial median of each component,

$$\mathbf{f}^{\text{res}}_{j,i} = M\big(\mathcal{W}_{\boldsymbol{\delta}_{j,i-1}}\mathbf{x},\ \mathcal{W}_{\boldsymbol{\delta}_{j,i}}\mathbf{x}\big) - \operatorname{med}_{\mathbf{p}}\ M\big(\mathcal{W}_{\boldsymbol{\delta}_{j,i-1}}\mathbf{x},\ \mathcal{W}_{\boldsymbol{\delta}_{j,i}}\mathbf{x}\big)\,. \tag{4}$$

On the same test this leaves 0.33 pixels, a factor of 15 smaller. The median is used rather than the mean so that a strong local response does not shift the estimate of the global component.

There is also a perceptual reason to prefer (4). A uniform translation of the whole retinal image is what an eye movement produces, and observers do not perceive it as object motion. What is perceived is the part of the field that departs from uniform. The cost is that a genuinely uniform illusory drift across the entire frame would be invisible to the measurement. Every target we use is spatially structured, so this does not bind, but a whole-frame constant target would need the direct subtraction instead.

The ensemble average over frames and directions is

$$\bar{\mathbf{f}}(\mathbf{x}) = \frac{1}{J(K-1)} \sum_{j=1}^{J} \sum_{i=2}^{K} \mathbf{f}^{\text{res}}_{j,i}. \tag{5}$$

When $K = 1$ or $\Delta = 0$ this is zero for every image, which is the empty objective from the introduction, and is the ablation reported in the experiments.

One consequence of (5) is worth stating because it affects how the ablations should be read. When the J directions are spread evenly, the offsets sum to approximately zero, so the gain error described above also cancels in the average. The direct subtraction and (4) therefore look similar once J is large, and they differ sharply only at $J = 1$. Diagnostics comparing the two must use a single direction or they measure nothing.

3.5 Score, loss, and guidance

The score is the masked normalised correlation used by Rosario et al. (2026),

$$\rho(\bar{\mathbf{f}}, \mathbf{F}^*) = \frac{\sum_{\mathbf{p}} m(\mathbf{p})^2\, \langle \bar{\mathbf{f}}(\mathbf{p}), \mathbf{F}^*(\mathbf{p}) \rangle}{\|\bar{\mathbf{f}}\|_\Omega\, \|\mathbf{F}^*\|_\Omega + \varepsilon} \ \in [-1, 1]. \tag{6}$$

By Cauchy–Schwarz $\rho = 1$ exactly when $\bar{\mathbf{f}} \propto \mathbf{F}^*$ with a positive constant on Ω; $\rho(\mathbf{F}, -\mathbf{F}) = -1$; and ρ vanishes for fields orthogonal at every pixel, such as a rotation field against its own radial counterpart. We verified all three numerically to within 10^{-3}.

The full objective adds a content term and, for stencil targets, a quiet term,

$$\mathcal{L}(\mathbf{x}) = \underbrace{\lambda_m\big(1 - \rho(\bar{\mathbf{f}}(\mathbf{x}), \mathbf{F}^*)\big)}_{\text{motion}} + \underbrace{\lambda_c\big(1 - \cos(E_I(\mathbf{x}), E_T(\text{prompt}))\big)}_{\text{content}} + \underbrace{\lambda_q\, Q(\bar{\mathbf{f}}(\mathbf{x}))}_{\text{quiet}}, \tag{7}$$

where E_I, E_T are frozen CLIP image and text encoders Radford et al. (2021) and

$$Q(\mathbf{f}) = \frac{\sum_{\mathbf{p}} \big(1 - m(\mathbf{p})\big)^2 \|\mathbf{f}(\mathbf{p})\|^2}{2 \sum_{\mathbf{p}} \big(1 - m(\mathbf{p})\big) + \varepsilon} \tag{8}$$

penalises residual flow outside the support. Without Q a stencil target tends to make the whole frame shimmer, and "only the word moves" is not legible.

Algorithm 1 Jitter-ensemble motion guidance (one sample)

Require: target $\mathbf{F}^*$, mask m, prompt c, frozen generator and reader M, offsets $\{\boldsymbol{\delta}_{j,i}\}$, weights $\lambda_m, \lambda_c, \lambda_q$

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \dots, 1$ **do**
  $\hat{\mathbf{x}}_0 \leftarrow$ Tweedie estimate (9)
  if t in guidance window **then**
    $\bar{\mathbf{f}} \leftarrow$ ensemble residual (4)–(5) evaluated at $\hat{\mathbf{x}}_0$
    $\mathcal{L} \leftarrow \lambda_m(1 - \rho(\bar{\mathbf{f}}, \mathbf{F}^*)) + \lambda_c \mathcal{L}_{\text{content}} + \lambda_q Q(\bar{\mathbf{f}})$
    $\mathbf{x}_t \leftarrow \mathbf{x}_t - s_t \, \Pi_\tau[\nabla_{\mathbf{x}_t} \mathcal{L}]$
  end if
  $\mathbf{x}_{t-1} \leftarrow \textsc{SamplerStep}(\mathbf{x}_t, \epsilon_\theta, t)$
end for
return $\mathbf{x}_0$

Guidance. For a diffusion generator with noise prediction ϵ_θ and cumulative noise schedule $\bar{\alpha}_t$, the current estimate of the clean image is the standard Tweedie form

$$\hat{\mathbf{x}}_0(\mathbf{x}_t, t) = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\, \epsilon_\theta(\mathbf{x}_t, t, c)}{\sqrt{\bar{\alpha}_t}}, \tag{9}$$

which follows from the forward process $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\, \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$ Ho et al. (2020). We evaluate (7) at $\hat{\mathbf{x}}_0$ and take a step against its gradient in the state, following Dhariwal and Nichol (2021); Bansal et al. (2023),

$$\mathbf{x}_t \;\leftarrow\; \mathbf{x}_t - s_t \, \Pi_\tau \big[\nabla_{\mathbf{x}_t} \mathcal{L}\big(\hat{\mathbf{x}}_0(\mathbf{x}_t, t)\big) \big] , \tag{10}$$

with Π_τ a norm clip at τ and s_t a decaying scale. Gradients pass through the reader and the warps only; ϵ_θ is treated as constant within the step, so no weights are touched and no graph is kept through the sampler. For a latent model the decoder sits between $\hat{\mathbf{x}}_0$ and the loss and is likewise frozen and differentiated through Rombach et al. (2022); Podell et al. (2024). Guidance is applied on an interior window of steps: early states are too noisy for a motion reader to say anything useful, and late steps mainly add texture.

Because (7) is a scalar function of the image, the same objective drives gradient-free optimisers when gradients are unavailable: best-of-N over initial noise Ma et al. (2025), or particle steering Singhal et al. (2025). Only the optimiser changes; the objective and the evaluation are the same.

3.6 A reader that needs no download

Our in-loop reader is intended to be the dual-channel model of Sun et al. (2025), the one model reported by Rosario et al. (2026) to show the human direction under simulated saccades. Because its weights are gated and its gradient behaviour cannot be checked in advance, we also supply a small reader with no learned parameters, so the method is runnable and testable on its own. It has three stages.

Latency. Elements at different luminances are registered at slightly different times Conway et al. (2005); Backus and Oruç (2005), and a temporal filter of this kind is the mechanism proposed by Fermüller et al. (2010). We apply a one-tap temporal blend whose weight depends on local luminance L,

$$I_t^{\text{eff}} = (1 - a)\, I_t + a\, I_{t-1}, \qquad a = \gamma_{\text{lat}}\,(1 - L), \tag{11}$$

so darker regions lag. This stage is necessary rather than decorative: with $\gamma_{\text{lat}} = 0$ the reader measures $\rho = +0.012$ on a peripheral-drift pattern at 192×192, against $+0.100$ with $\gamma_{\text{lat}} = 0.75$. Without it there is no illusion to detect.

Quadrature filtering and phase velocity. For orientation $\mathbf{n}_\phi = (\cos\phi, \sin\phi)$ and spatial frequency f, we build a Gabor pair with a Gaussian envelope g,

$$G_e(\mathbf{u}) = g(\mathbf{u}) \cos(2\pi f\, \mathbf{u}\cdot\mathbf{n}_\phi)\,, \qquad G_o(\mathbf{u}) = g(\mathbf{u}) \sin(2\pi f\, \mathbf{u}\cdot\mathbf{n}_\phi)\,, \tag{12}$$

each with its mean subtracted so a flat region gives no response. Writing e_t, o_t for the responses and $\varphi_t = \operatorname{atan2}(o_t, e_t)$, the component velocity along $\mathbf{n}_\phi$ follows the phase-based construction of Fleet and Jepson (1990). For a grating $I(\mathbf{p}) = \cos(2\pi f\, \mathbf{p}\mathbf{n}_\phi)$ shifted by $\boldsymbol{\delta}$, equation (3) gives $I(\mathbf{p}-\boldsymbol{\delta}) = \cos(2\pi f\, \mathbf{p}\mathbf{n}_\phi - 2\pi f\, \boldsymbol{\delta}\mathbf{n}_\phi)$, so the true phase changes by $-2\pi f(\boldsymbol{\delta}\cdot\mathbf{n}_\phi)$. The convolution in standard deep-learning libraries is cross-correlation, which negates the measured phase, and the recovered velocity is therefore

$$v_\phi = \frac{\Delta\varphi}{2\pi f}, \qquad \Delta\varphi = \operatorname{atan2}\big(\sin(\varphi_1 - \varphi_0),\, \cos(\varphi_1 - \varphi_0)\big), \tag{13}$$

with the atan2 form wrapping the difference to $(-\pi, \pi]$ differentiably. The sign in (13) is easy to get wrong, and getting it wrong reverses every result while leaving all of them plausible; we test it directly by shifting a texture by a known amount.

Combination. Each orientation constrains only the velocity component along $\mathbf{n}_\phi$, the aperture problem. We combine the constraints by weighted least squares, with weights $w_\phi = A_0 A_1$ given by the product of the quadrature amplitudes, and Tikhonov damping λ so that untextured regions do not produce large velocities:

$$\mathbf{A} = \sum_{\phi,f} w_\phi\, \mathbf{n}_\phi \mathbf{n}_\phi^\top, \quad \mathbf{b} = \sum_{\phi,f} w_\phi\, v_\phi\, \mathbf{n}_\phi, \quad \mathbf{v} = (\mathbf{A} + \lambda \mathbf{I})^{-1}\mathbf{b}. \tag{14}$$

This is the standard intersection-of-constraints solution; the recurrent attention stage of Sun et al. (2025) plays the corresponding role in the model we use when it is available.

An honest gap. Equation (11) makes lag decrease with luminance. Conway et al. (2005) instead report that response latency depends on *contrast relative to the mean*: black and white elements peak 10–20 ms before the intermediate greys. We implemented that version as well. It produces a stronger separation between a drift pattern and its luminance-reversed twin (0.28 against 0.20 in ρ), but with the opposite sign. We do not resolve which is right here; both are simplifications of a mechanism that is still debated Conway et al. (2005); Backus and Oruç (2005); Fermüller et al. (2010). The practical consequence is that a reader's direction convention must be calibrated once against a stimulus whose human direction is known, and that no claim in this paper rests on a single reader. It is also the clearest argument we can give for the protocol in Section 3.7.

3.7 Controls

Three checks are built into the evaluation.

Held-out readers. The reader used in (7) never appears in reported numbers. Reported scores come from readers of a different family Wang et al. (2024); Lotter et al. (2017), which are never differentiated through. This is enforced in code rather than by convention.

Signed control. Flipping ω in (1) must flip the sign of the measured ρ. A method that produces motion energy without controlling its direction passes the unsigned test and fails this one.

Negative control. A transform that removes the percept for a human viewer must drive ρ toward zero. Our first attempt was a single global permutation of the quantised luminance levels, which preserves the layout and the set of luminances present. It does not work, and the way it fails is informative: a global permutation still leaves a coherent asymmetric sequence, just a different one, so it yields a different illusion rather than none. The measured ρ rose from $+0.100$ to $+0.390$. What works is permuting the levels independently within local blocks, which keeps the local histogram and the contrast but destroys the ordering across the pattern. We retain the global permutation as a *positive* control, where a high score is the expected outcome.

3.8 Stimuli and reproducibility

Calibration and initialisation use procedurally generated drift patterns rather than published illusions, which are copyrighted. Elements are laid out in concentric rings with the asymmetric luminance sequence of Kitaoka and Ashida (2003), at an angular repeat count that scales with resolution. The last point matters: with a fixed repeat count the pattern aliases at small sizes and the illusion disappears without warning. Measured

ρ on the same pattern rendered at three sizes was +0.003 at 96 pixels, +0.051 at 128, and +0.100 at 192, with the jitter step scaled with resolution. Since a stage-I pixel-space diffusion model generates at 64×64, this determines where guidance can be applied, and we treat the guidance stage as a setting to be checked rather than assumed.

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